

SHELJIN S.A

IoT & Web Solutions Engineer

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PROFESSIONAL SUMMARY

Expert in connected devices, embedded systems, MQTT, cloud IoT platforms, and sensor integration, delivering secure, low-latency, scalable, fault-tolerant, and high-performance solutions. Skilled in real-time data analytics, edge computing, device-to-cloud communication, ROS 2 integration, Celery, Redis, AI/ML algorithms, and optimizing device performance for diverse industrial and consumer applications.

WORK EXPERIENCE

Embedded Engineer | Foxtech Private Solutions

03/2025 - Present

AI Rover

- Built an autonomous robotic system using Raspberry Pi 5, running two AI models concurrently for object detection and classification, integrating sensor controls and optimized with multi-threading for real-time performance.
- Implemented multiple autonomous modes with real-time live camera feedback, IMU sensor integration, and compass data for precise orientation, supporting full offline operation.
- Managed device control and sensor data using threading and FastAPI endpoints with async/await, achieving sub-100 ms response latency.
- Optimized navigation algorithms to improve obstacle avoidance by 30%, ensuring seamless hardware-software integration with motors, sensors, and IoT modules.

ROS 2 Mapping

- Developed a 3D environment mapping system using LiDAR sensors and ROS 2, creating real-time point cloud models for autonomous navigation and obstacle detection.
- Implemented SLAM algorithms for precise localization and 3D reconstruction, achieving sub-5 cm spatial accuracy.
- Simulated and tested in Gazebo, optimizing data processing pipelines for low-latency mapping and scalable integration with robotics platforms.

Python/Django Developer | Clovion Tech Solutions Pvt., Ltd.

05/2024 - 02/2025

E-Commerce API

- Designed and implemented scalable RESTful APIs using DRF, Swagger UI, Celery, Redis, 2FA, JWT, and Eraser.io.
- Optimized backend performance by 25%, improving system stability and response times.
- Collaborated within a 5-member agile team, focusing on microservices, asynchronous tasks, and real-time API design using WebSockets.

Billing Software

- Built a high-performance billing system using Django, ReportLab, and Celery for automated invoice generation.
- Integrated MySQL with ORM and Redis caching to boost query efficiency by 30%.
- Focused on deployment, CI/CD, and reliability using GitHub Actions and cPanel, ensuring version consistency across multiple environments.

Hardware Deployment & Management

- Deployed DRF APIs, IoT MQTT brokers, and static web apps across cPanel and cloud environments.
- Managed MySQL and Redis databases, automated background tasks using Celery, and reduced REST API latency by 20%.
- Maintained GitHub-based version control and streamlined CI/CD pipelines for scalable, production-ready systems.

PERSONAL PROJECTS

Sound or Mode Controlled - <https://github.com/Shelji/nano-soundcontrol-pwm-manager>

- Developed a sound or mode-controlled automation system using Arduino Nano, PWM, and utime/microtime methods for precise signal timing.
- Integrated a microphone sensor and multiple lighting control modes, achieving a 25% improvement in reaction accuracy and enhanced real-time responsiveness through optimized interrupt handling and hardware-level tuning.

Bike Locker - https://github.com/Sheljji/bike_lock

- Developed an IoT-based smart bike locker system using Arduino and a Bluetooth module for secure, real-time lock/unlock operations.
- Implemented mobile control to enhance user interaction and improved response time by 30% through optimized asynchronous communication between hardware and server, ensuring reliability and scalability.

Web Scrapper Image Download - <https://web-scrapper-image-downloader.streamlit.app/>

- Built with Python, Streamlit, and Playwright to extract 1000+ high-quality images from Unsplash, including premium, watermark-free content in maximum resolution via automated scraping.

Facial Attendance System - <https://github.com/Sheljji/admin-panel>

- Developed a real-time facial recognition attendance system using Django, OpenCV, and NumPy, integrating IoT cameras for live authentication and automated record logging.
- Enhanced system accuracy by 15% through optimized image preprocessing and implemented secure database management for reliable attendance tracking.

Battery Management - https://github.com/Sheljji/battery_manager

Plant-Moisture Data Analyzer - https://github.com/Sheljji/-raspberry_pi_pico_w_Thingspeak_DHT

E-Commerce Website - <https://shelj.pythonanywhere.com/>

Gamers World - <https://gamer.shelj.in/>

SKILLS

Programming Languages: Python, JavaScript, HTML, CSS, C++, PHP, SQL (MySQL, PostgreSQL), Bash/Shell

Wired & Wireless Technologies: I2C, ADC, SPI, UART, CAN, Ethernet, Wi-Fi, Bluetooth, RS-232

Networking Technologies: TCP/IP, DNS, DHCP, HTTP, MQTT, SSL/TLS

Development Boards & Embedded Systems: Micro Python, Arduino, Raspberry Pi, ESP8266, Sensors (IMU, LiDAR), IoT Devices

Backend: Django, Django REST Framework (DRF), FastAPI, Celery, MQTT, Redis

Frontend: React.js, HTMX, Tailwind CSS, Bootstrap, GSAP, Jquery

AI & Computer Vision: YOLO, OpenCV, NumPy, Pandas

Automation & Tools: Selenium, ReportLab, Eraser.io

API & Documentation: RESTful APIs, Swagger UI, Postman

OS & Cloud: Amazon Web Services, Linux, Raspbian, Windows, cPanel, PythonAnywhere, Hostinger, Netlify, IoT Cloud Platforms (Thingspeak, MQTT brokers, Blynk)

Tools & Version Control: Git, GitHub, CI/CD pipelines, Version Control, Streamlit

Technologies & Concepts: Microservices Architecture, Web Scraping, Asynchronous Programming (async/await, threading), Edge Computing, IoT Integration, Real-time Data Analytics, ROS 2, MQTT, Sensor Fusion

COMPETITIVE CODING PROFILE

- **CodeChef** - codechef.com/users/sheljji
- **Leetcode** - leetcode.com/u/sheljji/

Certificates

- **Python/Django** - <https://shelj.in/certificates/>
- **Mechatronics - Acutro Technologies Pvt Ltd (India, Germany)** - <https://shelj.in/certificates/>
- **National Cadet Corps** - <https://shelj.in/certificates/>
- **Competitions** - <https://shelj.in/certificates/>

EDUCATION

Bachelor of Engineering in Electrical and Electronics

05/2020 - 06/2024

Mar Ephraem College of Engineering and Technology

Elavuvilai

INTERESTS

- Surf Tech on YouTube, Football, IOT, Problem solving, Electrical, Craft works, Servers (NAS)